

This project is about analyzing the IPL 2022 auction data. The main goal was to understand how teams bought players and find useful insights from the auction. First, I loaded the dataset and cleaned the data by removing unnecessary columns and handling missing values. Then I created some new columns, such as whether a player was sold or unsold. After that, I analyzed the data and created graphs to see the distribution of player types, the number of players bought by each team, sold versus unsold players, and retained versus auction players. I used Python, Pandas, Matplotlib, and Seaborn for this project. Through this project, I learned data cleaning, data analysis, and data visualization."

What exactly did you do?, you can say:

"I cleaned the dataset, handled missing values, created new features, grouped and filtered data using Pandas, and generated charts to understand player and team statistics. Finally, I identified insights such as top bowlers by auction price and players who played in IPL 2021 but remained unsold in IPL 2022."

-----start-----

This project is an Exploratory Data Analysis (EDA) project on the IPL 2022 Auction dataset. The main objective was to clean the data, analyze player and team statistics, and generate visual insights from the auction data.

First, I imported the required libraries. I used Pandas for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for data visualization. I also suppressed warning messages to keep the output clean.

Next, I loaded the IPL 2022 dataset into a Pandas DataFrame using `read_csv()`. After loading the dataset, I explored its structure using `shape`, `info()`, and `columns`. The `shape` function gives the number of rows and columns, `info()` provides column names, data types, and non-null counts, and `columns` displays all column names.

Then I performed data cleaning. I removed an unnecessary column called 'Unnamed: 0', which was just an index column. After that, I checked for missing values using `isnull().sum()`.

I found missing values in the player cost columns and replaced those NaN values with 0 using `fillna()`. This indicates that those players were either unsold or had no auction price. Similarly, I found missing values in the '2021 Squad' column and replaced them with 'Not Participated' so that the dataset became more meaningful and easier to analyze.

After cleaning the data, I created a new feature called 'status'. First, I collected all teams whose players had a cost greater than zero. These represent players who were sold in the auction. Then I replaced those team names with the value 'sold' in the new status column. This helped me classify players as sold or unsold.

Next, I checked for duplicate player names using `duplicated()`. This was done to identify players whose names appeared more than once in the dataset.

I then analyzed player categories using the TYPE column. Using `value_counts()`, I counted how many players belonged to each category such as Batsman, Bowler, All-Rounder, and Wicket Keeper. After converting the result into a DataFrame using `reset_index()`, I visualized the distribution using a pie chart. This chart showed the percentage contribution of each player category.

To understand auction outcomes, I created a bar chart showing the number of sold and unsold players. I used Seaborn's `countplot()` and displayed the exact count on top of each bar using `annotations`. This provided a clear comparison between sold and unsold players.

I also calculated the total number of sold and unsold players using `groupby()` and `count()`. This gave a numerical summary of auction results.

Next, I analyzed team-wise purchases. I filtered out the unsold players and created a bar chart showing how many players each team bought during the auction. I again displayed the exact player counts on top of each bar for better readability.

After that, I created a new column called 'retention' from the Base Price column. The purpose was to distinguish retained players from players purchased during the auction. I replaced regular auction base prices such as '2 Cr', '1 Cr', and '50 Lakh' with the value 'From Auction'. This allowed me to categorize players as either retained or auction-bought.

I also handled the 'Draft Pick' value by replacing it with 0 to simplify further processing.

Then I performed feature engineering on the Base Price column. I created two new columns: 'base\_price\_unit' and 'base\_price'. The `base_price_unit` column stores units such as 'Cr' or 'Lakh', while the `base_price` column stores only the numeric value. This separation makes future analysis easier.

To understand squad composition, I grouped the data by Team and Retention status. This helped identify how many players each team retained and how many were bought through the auction.

I then created another visualization using `countplot()` with the TYPE column as hue. This chart showed the number of Batsmen, Bowlers, All-Rounders, and Wicket Keepers in each team, helping analyze team balance and squad structure.

I also planned to analyze the highest amount spent by each team on a single player and identify the retained player with the highest retention value. These analyses were included as comments for future enhancement.

Next, I identified the top five bowlers based on auction price. I filtered only bowlers, sorted them in descending order based on their cost, and selected the top five using `head(5)`. This helped identify the most expensive bowlers in the IPL 2022 auction.

Finally, I renamed the '2021 Squad' column to 'Prev\_team' for better readability. Using this column, I identified players who participated in IPL 2021 but went unsold in IPL 2022. I filtered players whose previous team was not 'Not Participated' and whose current team was 'Unsold'. This provided a list of experienced players who did not get selected in the auction.

Overall, this project helped me learn data cleaning, handling missing values, feature engineering, grouping and filtering data, and creating meaningful visualizations using Pandas, Matplotlib, and Seaborn.